AFC–7763

Sub. Code
1MPH4E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fourth Semester

Physics

***Elective* — MEDICAL PHYSICS**

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write briefly the function of breathing.

2. What is the function of electro cardiogram?

3. Write briefly the properties of ultra sound.

4. Write the function of stethoscope.

5. Write the units of light.

6. Write the use of infra red light in medicine.

7. How X-ray is generated?

8. Write the applications of X-ray in diagnosis.

9. Write the diastolic and systolic readings in normal and high blood pressure.

10. Mention few cardiovascular diseases.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Discuss the function of electroencephalogram.

Or

(b) Write the measurement of lung volume.

12. (a) Describe the ultra sound images of the body.

Or

(b) Describe the use of ultra sound to measure the motion.

13. (a) Define laser. Describe the applications of lasers in medicine.

Or

(b) Write the important components of a light microscope. Discuss the application of in medicine.

14. (a) Discuss how X-ray image is formed.

Or

(b) Discuss about X-ray fluoroscopy and mention its use in diagnosis.

15. (a) Discuss major components of cardiovascular system.

Or

(b) Discuss Bernoulli's principle and blood flow.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss the magnetic signals from heart and brain.

17. Discuss the function of outer ear, middle ear and inner ear.

18. Write the application of ultra violet and infra red light in medicine.

19. Discuss the radiograph taken without the film.

20. Discuss the exchange of oxygen and carbon di-oxide in capillary system.

AFC–7752

Sub. Code
1MPH1E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015.

First Semester

Physics

Elective — NUMERICAL METHODS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. Write a note on errors of observation and measurement.

2. Define curve fitting.

3. State the order of convergence and convergence condition for Newton's Raphson method.

4. How is Gauss-Jordan method different from Gauss elimination method?

5. State Gregory-Newton backward interpolation formula.

6. What is the Lagranges formula to find "Y", if three sets of values $(x_0, y_0), (x_1, y_1)$ and (x_2, y_2) are given?

7. When can numerical differentiation be used?

8. State Trapezoidal rule to evaluate $\int_{x_0}^{x_n} y(x)dx$.

9. State the modified Euler's formula.

10. Mention the advantages of Runge-Kutta method.

Section B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Discuss the disadvantages of the method of averages when compared to the method of least squares.

Or

(b) Describe a method for fitting a straight line for a related set of data.

12. (a) Find the positive root of the following equation using Newton-Raphson method $x^3 - 5x - 6 = 0$.

Or

(b) Solve the following system of equation by Gauss-Seidal method.

$$\begin{aligned} 5x_1 - x_2 &= 9 \\ -x_1 + 5x_2 - x_3 &= 4 \\ -x_2 + 5x_3 &= -6. \end{aligned}$$

13. (a) Apply Newton's backward formula to find a polynomial of degree 3 which includes the following x, y pairs.

x	3	4	5	6
y	6	24	60	120

Or

(b) Derive the Gauss backward interpolation formula.

14. (a) Find the values of $f''(5), f''(3.3)$ from the following table using numerical differentiation based on Newton's backward interpolation formula.

x	0	1	2	3	4	5
$f(x)$	4930	5026	5122	5217	5312	5401

Or

(b) Describe the Trapezoidal method of computing integrals.

15. (a) Solve $\frac{dy}{dx} = 1 - y,\ y(0) = 0$, using Euler's method. Find y at $x = 0.1$ and $x = 0.2$.

Or

(b) Write a C program for solving ordinary differential equations using Runge-Kutta second order method.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. By the method of least squares fitting, find the best fitting straight line to the data given below.

x	5	10	15	20	25
y	15	19	23	26	30

17. Solve the following system of equations by Gauss elimination method.

$$3x + y - z = 3$$
$$2x - 8y + z = -5$$
$$x - 2y + 9z = 8.$$

18. (a) Write a C program to implement Lagrange's interpolation.

(b) Construct Newton's forward interpolation polynomial for the given data :

x	4	6	8	10
y	1	3	6	16

19. (a) Compute $\int_0^1 \frac{dx}{1+x^2}$ by using trapezoidal rule, taking $h = 0.5$ and $h = 0.25$.

(b) Write a C program to evaluate $\int_a^b f(x)dz$ numerically using Simpson's $\frac{1}{3}$ rule.

20. Find the value of $y(0.2)$ and $y(0.4)$ using Runge-Kutta method of the fourth order with $h = 0.2$, given that $\frac{dy}{dx} = \sqrt{x^2 + y}$; $y(0) = 0.8$.

AFC–7753

Sub. Code
1MPH2C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Physics

MATHEMATICAL PHYSICS — II

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** the questions.

1. Find the sine transforms of e^{-x}.

2. State convolution theorem in Fourier transform.

3. Find regular singular points of the differential equation :

$$2x^2 \frac{d^2y}{dx^2} + 3x \frac{dy}{dx} + (x^2 - 4)\, y = 0.$$

4. Classify the equation $2 \frac{\partial 2y}{\partial x^2} + 4 \frac{\partial^2 y}{\partial x \partial y} + 3 \frac{\partial^2 y}{\partial y^2} = 0$.

5. If A_{ij} is an antisymmetric tensor of second order and U' is a tensor of rank one then show that $A_{ij} u' u' = 0$.

6. What is invariant tensor? Give example.

7. State the orthogonality theorem.

8. What are point groups? Give an example.

9. Show that $H_n(-x) = (-1)^n H_n(x)$.

10. Prove that $\Gamma(n+1) = n\Gamma(n)$.

Part B $(5 \times 5 = 25)$

Answer **all** the questions, choosing either (a) or (b).

11. (a) Find the inverse Laplace transform of $L^{-1}\left[\frac{s-3}{s^2+4s+3}\right]$.

Or

(b) Using Fourier sine and cosine transform evaluate

$$\int_0^\infty \frac{\cos nx}{a^2+n^2}\,dn \text{ and } \int_0^\infty \frac{n \sin nx}{a^2+n^2}\,dn.$$

12. (a) Solve the linear partial differential equation :

$$\frac{\partial^4 z}{\partial x^4} + \frac{\partial^4 z}{\partial y^4} = 0.$$

Or

(b) A string of length l is vibrating with frequency w. The equations and boundary conditions are $\frac{d^2y}{dx^2} + k^2 y = 0$, $u(0) = 0 = u(l)$, $k = \frac{w}{c}$. Solve the problem.

13. (a) Show that Kronecker delta δ_V^H is a mixed tensor and has the same components in every coordinate system.

Or

(b) Discuss the application of tensor in moment of inertia.

14. (a) Prove that the group of order 2 and 3 are always cyclic but a group of order 4 may (or) may not be a cyclic group.

Or

(b) Prove that two right cosets of a subgroup in a given group are either equal (or) else no elements in common.

15. (a) Show that $\int_{-1}^{+1} xP_n(x)P_{n-1}(x)\,dx = \frac{2n}{4n^2 - 1}$.

Or

(b) Write a note on generating function for $J_n(x)$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Obtain Laplace transforms of the function $F(t) = \sin h\, at \sin at$.

17. Using separation of variables method derives the general solution of the ware equation of the vibrating string fastened at two ends to find supports

$\frac{d^2y}{dx^2} = \frac{1}{c^2}\frac{d^2y}{dt^2}$ with $y(x,t) = y_0(x)$, $\frac{dy}{dt} = V_0(x)$ at $t = 0$.

18. Explain covariant, contravariant and mixed tensors with example.

19. (a) Explain character table for C_{3V} point group.

(b) State and prove Schur's Lemma 2.

20. Obtain the solution of the Hermite differential equation :

$\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2xy = 0$ in the form of power series.

AFC–7754

Sub. Code
1MPH2C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015.

Second Semester

Physics

ELECTROMAGNETIC THEORY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. Write down the Poisson's equation and Laplace equation related to the electric field.

2. State the Coulomb's law.

3. State the Lorentz force law.

4. What is Ampere's law of force?

5. What is displacement current?

6. State Faraday's laws of electromagnetic induction.

7. What do you mean by plane wave?

8. Define 'Plasma'.

9. Define : Magnetic dipole.

10. What is meant by retarded potential?

Section B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Discuss the classical Image problem.

Or

(b) How do you find the energy of a point charge distribution? Explain.

12. (a) State and prove Biot – Savart law.

Or

(b) Find out the divergence and curl of $\overline{B}$.

13. (a) Obtain the Maxwell's equation in a matter.

Or

(b) Obtain the expression for Poynting vector.

14. (a) Write notes on linear and circular polarization.

Or

(b) Explain the phenomenon of polarization by reflection and total internal reflection.

15. (a) What do you mean by scattering cross-section and also explain why does the sky appear blue?

Or

(b) Explain the multipole expansion for localized source.

Section C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. Establish Gauss's theorem for an electrostatic field. Deduce Laplace's and Poisson's equations.

17. (a) Write short notes on :

    (i) Magnetic susceptibility

    (ii) Magnetic permeability.

    (b) A current of 1A is flowing in the sides of an equilateral triangle of side 4.5×10^{-2} m. Find the magnetic field at the centroid of the triangle.

18. State and explain Poynting theorem.

19. Discuss the reflection and refraction of Electromagnetic waves at a plane interface between dielectrics.

20. Explain Rayleigh scattering on the basis of theory of scattering of electromagnetic waves.

AFC–7755

Sub. Code
1MPH2C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Physics

QUANTUM MECHANICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. Define expectation value.
2. What is meant by Hilbert Space?
3. Define Zero point energy.
4. If a rigid rotator is confined to rotate about a fixed axis represented by Eigen function $\varphi(\phi) = Ae^{im\phi}$ where $m = 0, \pm 1, \pm 2, ...$ Find its normalized Eigenfunction.
5. Define Rayleigh ratio.
6. What is meant by harmonic perturbation?
7. Write down the spin mathrices of spin half particles.
8. Define Partial waves.
9. What is meant by zitterbewegung?
10. Write down the Klein – Gordon equation for a free particle.

Section B (5 × 5 = 25)

Answer **all** questions by choosing either (a) or (b).

11. (a) Obtain equation of continuity for probability.

Or

(b) State and explain any two fundamental postulates of wave mechanics.

12. (a) Illustrate linear harmonic oscillator using operator method.

Or

(b) Explain rigid rotator in detail.

13. (a) Obtain Fermi's Golden Rule.

Or

(b) Describe sudden approximation in detail.

14. (a) Write a short note on scattering cross section.

Or

(b) Illustrate matrix representation of angular momentum in detail.

15. (a) Discuss Dirac Matrices in detail.

Or

(b) Write a note on negative energy state.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Show that the eigen values of Hermitian operators are real. (3)

(b) Show that any two eigen functions of a Hermitian operator that belong to different eigen values are orthogonal. (3)

(c) If A and B are Hermitian operators, show that (AB+BA) is Hermitian and (AB-BA) is not Hermitian. (4)

17. (a) Elaborate the Hydrogen atom using Schrodinger equation and obtain its wave functions. (6)

(b) For the ground state of the hydrogen atom, find the expectation value of the radius vector r of the electron. (4)

18. (a) Explain Zeeman Effect in detail. (6)

(b) Work out the splitting of the ${}^1p \rightarrow {}^1S$ transition of an atom placed in a magnetic field B along Z axis. (4)

19. (a) Discuss the addition of angular momenta. (6)

(b) The vector J gives the sum of angular moment J_1 and J_2. Prove that $(J_x, J_y) = i\hbar J_z$ .Also check whether $J_1 - J_2$ is an angular momentum or not. (4)

20. (a) Obtain Dirac equation for a free particle. (6)

(b) If $j(r,t) = c\psi^* \alpha \psi$ where Ψ is the four component wave vector and α's are Dirac matrices find j_x, j_y and j_z interms of four component wave vector. (4)

AFC–7756

Sub. Code
1MPH2E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Physics

Elective — NANO SCIENCE

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are nanostructures?

2. What are top–down nanofabrication?

3. What is plasma arching?

4. Mention the applications of nanomaterials.

5. Enumerate the types of nanotubes.

6. List out the different composition of each poker chip in DNA.

7. What are nano holes?

8. What is imaging?

9. What is nanoelectronics?

10. What are MEMS?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Describe the different types of nanotechnology.

Or

(b) Explain Top–down and bottom–up approach.

12. (a) Explain the chemical vapour deposition.

Or

(b) Discuss the Ball milling method for the preparation of nanopowders.

13. (a) Discuss the purification of carbon tubes.

Or

(b) Write a note on lipids as nano bricks and mortar.

14. (a) Explain the interaction of light by the nanostructures of nanotechnology.

Or

(b) Briefly describe about photonic crystals.

15. (a) Describe about the tools of micro and nano fabrication.

Or

(b) Discuss the preparation of new low cost energy efficient windows.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write a note on

(a) Molecular and Atomic size.

(b) Surfaces and dimensional space.

17. Discuss the sol–gel method for the preparation of nanomaterials.

18. Describe the properties and uses of nanotubes.

19. (a) Explain the properties of light and nanotechnology.

    (b) Write a short note on photonic crystal.

20. Explain the experimental implementation of quantum computers.

AFC–7757

Sub. Code
1MPH3C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Physics

THERMODYNAMICS AND STATISTICAL PHYSICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define first order phase transition.

2. What is reversible process? Give an example for it.

3. Define mean free path.

4. What are the assumptions made to derive Boltzmann transport equation?

5. A particle is moving in three dimensions. How many coordinates required to specify the motion of the particle in phase-space? Write them explicitly.

6. What is an ensemble? Give any two ensembles which are widely used.

7. What are Fermions? Write down the postulates of Fermi-Dirac statistics.

8. What are the differences between classical and quantum statistics?

9. Define paramagnetic susceptibility.

10. What is an Fermi energy?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Calculate the entropy change in an irreversible process.

Or

(b) Derive Clausis-Clapeyron's equation applicable for first order phase transition.

12. (a) Derive Maxwell-Boltzmann distribution law from Boltzmann transport equation.

Or

(b) Using Boltzmann's H-function define entropy and the second law of thermodynamics of an ideal gas in equilibrium.

13. (a) State and prove equipartition theorem.

Or

(b) If the partition function for a perfect gas is given by $z = \frac{v}{h^3}(2\pi mkT)^{3/2}$. Calculate (i) average kinetic energy per molecule (ii) specific heat of the gas, (iii) Helmholtz free energy, (iv) entropy and (v) enthalpy.

14. (a) Derive Bose-Einstein distribution law.

Or

(b) Distinguish between Classical statistics, Fermi-Dirac statistics and Bose- Einstein statistics.

15. (a) Discuss the application of B-E statistics to liquid Helium II.

Or

(b) Obtain Planck's formula for the distribution of energy in the spectrum of a black body radiation from statistical consideration applied to an assembly of photons.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Show that during the second order phase transition $\frac{\partial^2 G_1}{\partial P^2} - \frac{\partial^2 G}{\partial T^2} \neq 0$ and $\frac{\partial^2 G_1}{\partial P^2} - \frac{\partial^2 G_2}{\partial P^2} \neq 0$, Where G_1 and G_2 are Gibbs functions for phases 1 and 2 respectively.

(b) For helium gas $T_c = 5.0K$ and $p_c = 2.3\ atm$. Calculate the van der Waal's constants. Given $R = 8.314\,JK^{-1}mol^{-1}$ and $1\,atm = 1.013 \times 10^5\ Nm^{-2}$.

17. (a) Obtain Boltzmann transport equation explaining all the approximations made.

(b) Derive conversation of mass from Boltzmann transport equation.

18. (a) Obtain the partition function in the Grand canonical ensemble and deduce the thermodynamic: quantities there from.

(b) The probability function that a molecule can have its velocity component v_x along x-axis is $P(v_x) = \left(\frac{m}{2\pi kT}\right)^{\frac{1}{2}} e^{\frac{mv_x^2}{2kT}}$. Find the value of v_x for which the probability falls to $\frac{1}{e}$.

19. (a) Show that the specific heat of a strongly degenerate electron gas is directly proportional to its absolute temperature.

(b) Show that Bose – Einstein condensation does not occur in 2 dimensions.

20. (a) Using Debye's procedure, deduce an expression for the specific heat of monoatomic solid at very low temperatures.

(b) The Fermi energy E_F for lithium is 4.72 eV at T=0. Calculate the number of conduction electrons per unit volume in lithium. Take $m_e = 9.31 \times 10^{-31} Kg$ and $h = 6.62 \times 10^{-34}\ Js$.

AFC–7758

Sub. Code
1MPH3C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015.

Third Semester

Physics

ATOMIC AND MOLECULAR PHYSICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Classify fermions and bosons-based on the nature of wave functions.

2. Explain the physical significance of quantum numbers.

3. What is Stark effect?

4. Write a short note on Vander Wall's interactions.

5. HCl molecule has a B value of 10.593 cm^{-1} and a centrifugal distortion constant D of 5.3×10^{-4} cm^{-1}. Calculate the vibrational frequency and force constant of the molecule.

6. Explain the term overtone in vibration spectra.

7. What is meant by the term molecular polarizability?

8. Explain the meaning of predissociation with suitable illustration.

9. What are the requirements of observing electron spin resonance?

10. What is chemical shift in NMR spectroscopy?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Derive the expression for electron energy based Bohr atom model.

Or

(b) Discuss Stern and Gerlach experiment. Explain its importance.

12. (a) Discuss Paschen-Back effect in one and two electron systems.

Or

(b) Give a detailed description on bonding in H_2 molecule.

13. (a) Explain rotational spectra of rigid diatomic molecules.

Or

(b) Write a summary of analysis by infrared techniques.

14. (a) Explain the rotational fine structure in vibrational Raman spectra.

Or

(b) Discuss rotational fine structure on electronic-vibration transitions.

15. (a) Explain the function of an ESR spectrometer with neat diagram.

Or

(b) Discuss the applications of NMR spectroscopy.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. (a) Discuss LS and JJ coupling schemes with suitable illustration.

(b) Give a detailed account on organization of elements in a periodic table.

17. (a) Write down the Hückel secular determinant for 1,3-butadiene.

(b) Draw a molecular orbital diagram showing the energies of the molecular orbitals.

18. (a) Discuss the spectrum of diatomic vibrating rotator.

(b) What are the different modes of vibrations in tri-atomic linear and non-linear molecules? Give suitable illustrations.

19. (a) Discuss rotational fine structure on electronic-vibration transitions.

(b) Explain the Fortrat diagram and its properties.

20. (a) Give an overview of ESR spectroscopy and its applications.

(b) Calculate the g factor when the resonance was observed at 0.34 MHz in the field of 9506.690 Tesla.

AFC–7759

Sub. Code
1MPH3C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Physics

SOLID STATE PHYSICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define point group.

2. Sketch the structure of Diamond.

3. What are phonons?

4. What is phonon momentum?

5. Define electrical conductivity.

6. State Hall effect in semiconductors.

7. What is meant by dielectric loss?

8. How orbital motion of electron in iron group ions is quenched?

9. Discuss anti-ferromagnetic order.

10. What is SQUID? Give any two applications of SQUID.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Describe in detail the three dimensional Bravais lattices.

Or

(b) Write a short note on Rotating crystal method.

12. (a) Explain about elastic constants of crystals.

Or

(b) Discuss the vibrations of mono atomic lattices.

13. (a) State and prove Bloch's theorem.

Or

(b) Briefly explain the De Haas van Alphen effect.

14. (a) Find an expression for the local electric field acting on an atom.

Or

(b) Explain the process of cooling by adiabatic demagnetization.

15. (a) What is meant by thermal excitation of magnons and obtain $T^{3/2}$ law?

Or

(b) Enumerate the important results of BCS theory.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Show that the reciprocal of a BCC is FCC.

(b) NaCl crystallizes in cubic structure. The density of NaCl is 2180 kg m^{-3}. Calculate distance between adjacent atoms.

17. (a) The dispersion relation of a diatomic chain is given by $m_1 m_2 \omega^4 - 2C(m_1 + m_2)\omega^2 + 2C^2(1 - 2\cos ka) = 0$, for $ka << 1$. Find the frequencies of the optic and acoustic modes.

(b) Explain inelastic scattering of neutron by phonons.

18. (a) What are Brillouin zones? How are they related the energy levels of an electron in a metal?

(b) The density of zinc is 7.13 gm/cm^3 and its atomic weight is 65.4. Calculate its Fermi energy. The effective mass of a free electron in zinc crystal is 7.7×10^{-31} kg and the Avogadro's number is $6.02 \times 10 \times 10^{23}$ atoms/gm-atom.

19. (a) Derive Clausius-Mossotti relation.

(b) What is Hund's rule? Obtain the basic level of Eu^{3+} with configuration $4f^6 5f^2 p^6$.

20. (a) Outline the theory of AC Josephson effect.

(b) Calculate the critical current and critical current density for a sample of lead of diameter 1 mm of 4.2 K. The critical temperature for lead is 7.18 K and critical magnetic field is 6.5×10^4 A/m.

AFC–7760

Sub. Code
1MPH3E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015.

Third Semester

Physics

***Elective* — ENERGY PHYSICS AND ENVIRONMENTAL SCIENCE**

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. What is meant by renewable energy?

2. Write down the principle of a fuel cell.

3. Mention the advantages of pyroheliometer.

4. What do you understand from loss coefficient?

5. Name the types of concentrators.

6. State the principle of evacuated tube collector.

7. Write down solar photovoltaic principles.

8. List out the types of solar cells.

9. When do we say the water is polluted?

10. Name the available control devices for air pollution.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Write down the salient features of oil as an energy resource.

Or

(b) Study the Indian scenario of energy conservation and future demand.

12. (a) How does heat transfer in solar thermal radiation.

Or

(b) Derive the energy balance equation.

13. (a) Describe a solar air heater.

Or

(b) Explain the role played by high temperature collectors.

14. (a) Discuss about solar cell characteristics.

Or

(b) Elaborate the Indian perspective in the fabrication of solar cell.

15. (a) Discuss about the source of water pollution.

Or

(b) Explain the purification process of air pollution.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write in detail about various renewable energy sources.

17. Explain the principle, construction and working of a flat plate collector.

18. Describe a solar water heating system.

19. Explain about various types of solar cells.

20. How do you purify the water pollution and monitor the level of pollution?

AFC–7761

Sub. Code
1MPH4C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fourth Semester

Physics

NUCLEAR AND PARTICLE PHYSICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Comment on the charge distribution of nucleus.

2. Show that nuclear forces are spin independent.

3. What is nuclear isomerism?

4. How does the charged particle interact with matter?

5. List out the various fission reactions

6. What is pinch effect in nuclear fusion?

7. Write down any two conservative laws of nuclear reactions.

8. (d,p) stripping reactions are more frequent than (d,n) reactions. Why?

9. Define strangeness.

10. Is the following reaction is charge conserved, $\pi^+ + n \rightarrow \pi^- + p$?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Write the semi-empirical mass formula and describe the importance of five constants involved in the formula.

Or

(b) Describe the fundamental and significance of Yukawa's meson theory.

12. (a) Classify beta transitions based on selection rules.

Or

(b) With neat diagram, explain the construction and working of Geiger-Muller counter.

13. (a) Draw the schematic diagram of betatron and explain its working.

Or

(b) Describe the various cycles involved in solar fusion.

14. (a) What is nuclear reaction kinematics? Comment on the significance of Q-equations.

Or

(b) Derive Breit Wigner dispersion formula.

15. (a) How are elementary particles classified?

Or

(b) Write a note on Gell-Mann Nishijima scheme.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. By explaining the vibrational and rotational states of nucleus, describe the importance of collective nuclear model.

17. Derive an expression for transmissivity of the barrier in Gamow's theory of alpha decay.

18. With neat diagram, explain Bohr and Wheeler's theory of nuclear fission.

19. Discuss the partial-wave analysis of reaction cross-section.

20. Describe Quark model in detail.

AFC–7762

Sub. Code
1MPH4C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fourth Semester

Physics

MICROPROCESSORS AND MICROCONTROLLER

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Name the addressing modes of 8085 microprocessor.
2. Give any two examples for logical instructions.
3. What is meant by synchronous data transfer?
4. Mention the principle of DMA controller.
5. What is a microcontroller?
6. What are the advantages of a microcontrollers?
7. How many instructions group are available in 8051? Name them.
8. Give the function of the following instruction.

   MOVX, A, @DPTR.
9. State the principle of a stepper motor.
10. What is meant by interfacing?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain the data transfer instructions.

Or

(b) Write an 8085 program to multiply two 8 bit numbers.

12. (a) Discuss about asynchronous data transfer.

Or

(b) Describe a programmable peripheral interface (8255 A).

13. (a) Write a note on microcontroller hardware.

Or

(b) Explain the function of timers in a microcontroller.

14. (a) Elaborate about bit manipulations.

Or

(b) Write an 8051 program to find the sum of N numbers.

15. (a) Briefly explain about the pulse measurement.

Or

(b) How do you interface a seven segment display?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. With suitable diagrams, describe the pins of 8085 and explain its architecture.

17. Explain about :

    (a) 8253 A (PIT)

    (b) 8259 (PIC)

18. Elaborate the memory structure of 8051.

19. Write an 8051 program to find the biggest and smallest numbers from a given list of numbers.

20. How do you apply 8051 for the conversion of one form of signal to another form?

AFC–7751

Sub. Code
1MPH1C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Physics

ELECTRONICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is tunnel diode?

2. How does the junction capacitance of a varactor diode vary with the variation in reverse bias voltage?

3. Draw the schematic symbols of n- channel and p- channel FET.

4. Mention the applications of TRIAC.

5. What are advantages of an active filter over a passive filter?

6. What is the concept of virtual ground in Op-Amp?

7. What are damped and undamped oscillations?

8. Calculate the oscillating frequency of an oscillator for the given values $L = 100\,\mu H$ and $C = 0.1\,\mu F$.

9. What is masking?

10. Mention the applications of monostable multivibrator.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Obtain the continuity equations of mobile carrier in any elementary volume of the semiconductor to prove the conservation of charge.

Or

(b) What is GUNN diode? Explain its characteristics with necessary circuit diagram.

12. (a) Draw the AC power control circuit using TRIAC. Explain its operation.

Or

(b) Describe the construction and working of JFET.

13. (a) Discuss the construction of inverting amplifier. Find its output voltage.

Or

(b) Explain the operation of low pass filter employing op-amp.

14. (a) Explain the construction and operation of sample and hold circuit using op-Amp.

Or

(b) Discuss the function of voltage control oscillator with suitable circuit diagram.

15. (a) Explain the method and fabrication of monolithic resistor.

Or

(b) Mention the merits and limitations of integrated circuits.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss the construction and working of LED. Bring out its applications.

17. Explain the construction and working of SCR. Discuss the V-I characteristics of SCR.

18. Sketch the circuit diagrams of integrator and differentiator using op-Amp. Obtain their output.

19. Illustrate the construction and working of triangular wave generator using op-Amp.

20. Describe the construction and working of monostable multivibrator using 555 timer.